Delhi's Air Pollution
Lab - 2

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1 The Question

How does NO2 concentration in Gujarat's port coastal districts compare with that in its industrial districts?

Globally, Ships contribute to around 15% of NOx levels. How would NOx levels around the Gujurat ports compare to those of its Industrial areas to see whether maritime activity may raise NOx levels around ports?[Yulnd]

2 Analysis

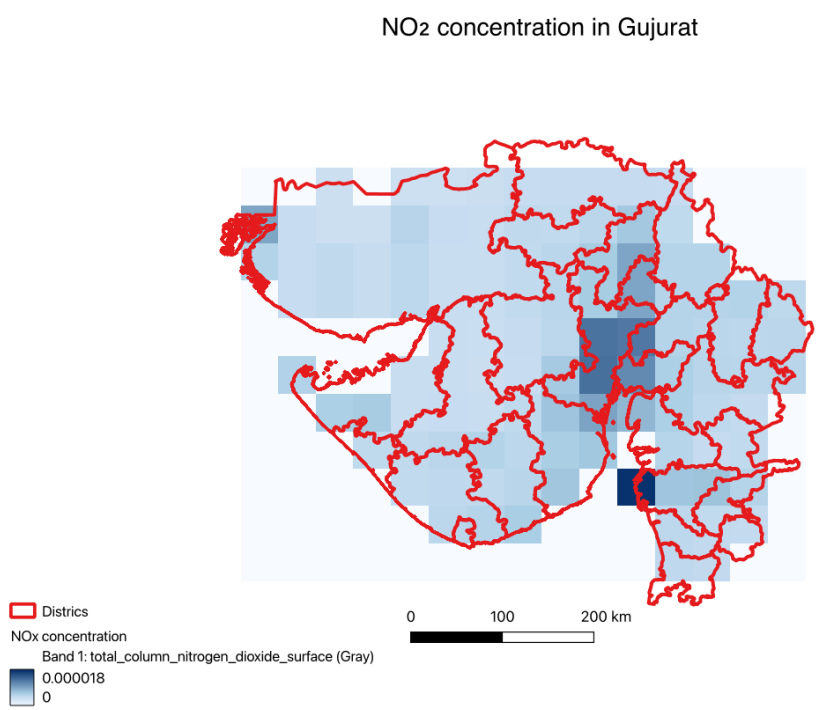


Figure 1: The following image shows the NO2 Concentration in the state of Gujurat with the districts highlighted

Refer to the political map of Gujurat Fig.2 for further clarity on the location of each district.

fid	District Name	censuscode	count	Sum	Mean
1	Ahmadabad	474	3	3.645701053756056e-05	1.215233684585352e-05
2	Amreli	480	5	2.846321604010882e-05	5.692643208021764e-06
3	Anand	482	2	1.787895780580584e-05	8.93947890290292e-06
4	Banas Kantha	469	6	2.876821054087486e-05	4.794701756812477e-06
5	Bharuch	488	3	1.861818145698635e-05	6.206060485662117e-06
6	Bhavnagar	481	4	2.777760164462961e-05	6.944400411157403e-06
7	Dohad	485	2	1.067340963345487e-05	5.336704816727433e-06
8	Gandhinagar	473	3	2.491860777809052e-05	8.30620259269684e-06
9	Jamnagar	477	6	3.255886076658498e-05	5.426476794430831e-06
10	Junagadh	479	4	2.131020664819516e-05	5.327551662048791e-06
11	Kachchh	468	24	0.0001181060579256155	4.921085746900644e-06
12	Kheda	483	2	1.86710476555163e-05	9.335523827758152e-06
13	Mahesana	471	3	2.01437387659098e-05	6.7145795886366e-06
14	Narmada	487	1.658765294098824	8.302047762889495e-06	5.00495627224938e-06
15	Navsari	490	1.291430502370883	6.372923308879705e-06	4.934778369552153e-06
16	Panch Mahals	484	3	1.703285943221999e-05	5.677619810739998e-06
17	Patan	470	3	1.603728287591366e-05	5.345760958637887e-06
18	Porbandar	478	1.342366477246646	5.113799707850618e-06	3.809540683956614e-06
19	Rajkot	476	8	3.839010605588555e-05	4.798763256985694e-06
20	Sabar Kantha	472	3	1.666708976699738e-05	5.555696588999126e-06
21	Surat	492	3	3.26246336044278e-05	1.08748778681426e-05
22	Surendranagar	475	7	3.648630718089407e-05	5.212329597270582e-06
23	The Dangs	489	1.023025184424825	3.88220577785952e-06	3.794829137116707e-06
24	Vadodara	486	7	3.864249356411165e-05	5.520356223444521e-06
25	Valsad	491	2	1.014051849779207e-05	5.070259248896036e-06
26	Tapi	493	2	1.072281338565517e-05	5.361406692827586e-06

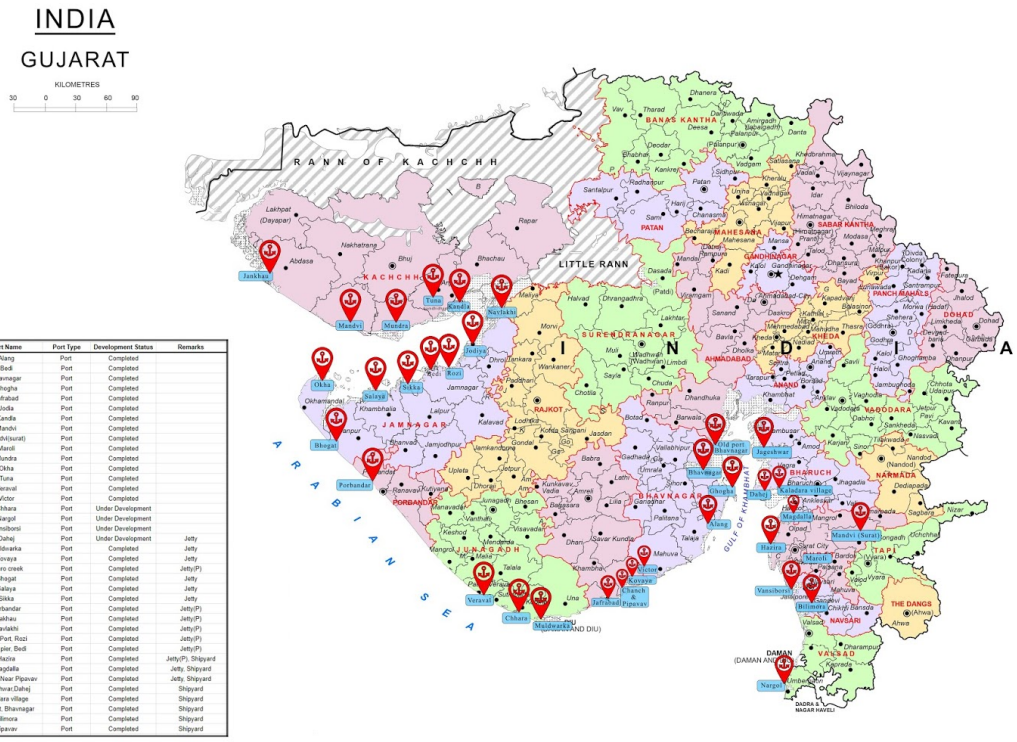
Table 1: The table shows the statistical obtained statistical values for each zone(district) in Gujurat

Based on the district-wise NO2 data, port coastal districts such as **Kachchh, Jamnagar, Porbandar, Bhavnagar, Surat, and Valsad** show moderate mean concentrations (1.08 - 4.9), while industrial districts like **Ahmedabad, Gandhinagar, Kheda, and Anand** exhibit higher values (around 6.2 and greater). This indicates that **industrial emissions contribute more strongly** to NO2 pollution than port-related shipping activities in Gujarat. With Ahmadabad having the highest.

3 Limitations and Advantages

The method facilitates comparative analysis of pollutant data across multiple districts and states, thereby enabling the identification of air pollution trends and the validation of theoretical predictions related to the influence of geographical factors or the distribution of specific emission sources, such as factories against observed values. A key limitation of the method lies in the resolution of the available data. While it is suitable for detecting patterns over relatively large geographical areas, it does not permit examination of pollution variability at finer spatial scales. For instance, one may ascertain that Ahmedabad is among the most polluted regions, but the method does not allow precise identification of specific high-pollution hotspots within the city. Consequently, the resolution is adequate for characterizing broad patterns but constrains the analysis of localized detail.

4 appendix



Serial No.	Port Name	Port Type	Development Status	Remarks
1	Alang	Port	Completed	
2	Bedi	Port	Completed	
3	Bhavnagar	Port	Completed	
4	Ghogha	Port	Completed	
5	Jafrabad	Port	Completed	
6	Jodia	Port	Completed	
7	Kandla	Port	Completed	
8	Mandvi	Port	Completed	
9	Mandvi(surat)	Port	Completed	
10	Maroli	Port	Completed	
11	Mundra	Port	Completed	
12	Okha	Port	Completed	
13	Tuna	Port	Completed	
14	Veraval	Port	Completed	
15	Victor	Port	Completed	
16	Chhara	Port	Under Development	
17	Nargol	Port	Under Development	
18	Vansiboosi	Port	Under Development	
19	Dahej	Port	Under Development	Jetty
20	Muldwarka	Port	Completed	Jetty
21	Kovaya	Port	Completed	Jetty
22	Kharo creek	Port	Completed	Jetty(P)
23	Bhagat	Port	Completed	Jetty
24	Salaya	Port	Completed	Jetty
25	Sikka	Port	Completed	Jetty
26	Porbandar	Port	Completed	Jetty(P)
27	Jakhau	Port	Completed	Jetty(P)
28	Navlakhi	Port	Completed	Jetty(P)
29	New Port, Rozi	Port	Completed	Jetty(P)
30	Rozi pier, Bedi	Port	Completed	Jetty(P)
31	Hazira	Port	Completed	Jetty(P), Shipyard
32	Magdalla	Port	Completed	Jetty, Shipyard
33	Chanch, Naar Pipavav	Port	Completed	Jetty, Shipyard
34	Jageshwar,Dahej	Port	Completed	Shipyard
35	Kaladara village	Port	Completed	Shipyard
36	Old Port, Bhavnagar	Port	Completed	Shipyard
37	Billimora	Port	Completed	Shipyard
38	Pipavav	Port	Completed	Shipyard

Figure 2: Political Map of Gujurat